**Haithem Turki**
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## Education

**Carnegie Mellon University,** Pittsburgh, PA — *2024 (expected)*
**PhD candidate in Computer Science**.
Thesis: Towards City-Scale Neural Rendering
Committee: Deva Ramanan (chair), Shubham Tulsiani, Jessica K. Hodgins, Angjoo Kanazawa, Jonathan T. Barron

**Stanford University,** Stanford, CA — *06/18*
**MS in Computer Science**.

**Stanford University,** Stanford, CA — *06/12*
**BS in Computer Science**, minor in Management Science and Engineering.

## Professional Experience

**Research Scientist Intern,** Meta, Pittsburgh, PA — *05/23-Present*
- Real-time neural rendering in VR with Drs. Christian Richardt and Michael Zollhoefer.

**Research Intern,** Argo AI, Pittsburgh, PA — *05/22-08/22*
- Explored city-scale 3D reconstruction methods with Professor Deva Ramanan and Dr. Francesco Ferroni

**Engineering Manager** and **Technical Lead,** Palantir Technologies, New York, NY — *08/12-08/19*
- Led R&D efforts to improve Palantir's Foundry data platform:
  - Created interactive search and analysis infrastructure that evolved into one of the primary user interfaces of Palantir's Foundry data lake platform deployed across the company's customer fleet
  - Invented full-text search system that powers Palantir's internal log analysis infrastructure and a significant part of the Foundry data lake platform
- Led inception and rollout of one of Palantir's flagship data analytics products
- Managed the technical implementation of a variety of high-profile customer engagements in the commercial sector (leading teams of 10-100+ individuals)

Supervised and mentored a team of 7+ direct reports

**Cofounder,** TradeSensei, Palo Alto, CA — *01/12-08/12*
- Founded a trading startup and wrote the backend infrastructure enabling beta users to participate in real-time trading competitions
- Participated in Stanford StartX accelerator program

**Software Engineering Intern,** Addepar, Mountain View, CA — *04/11-09/11*
- Worked on developing a software platform to help private financial advisors and family offices perform quantitative analysis and interface with their clients

**Research Assistant,** Stanford University, Stanford, CA — *06/10-12/10*
- Conducted computer vision research under Professor Daphne Koller
- Co-authored two accepted conference papers

## Teaching Experience

**Graduate Student Instructor, Dept. of Computer Science,** Carnegie Mellon University *09/20-Present*
- Teaching assistant for 15-640 (Distributed Systems) and inaugural 17-700 class (Data Science and Machine Learning at Scale)
- Supported and mentored team of undergraduate teaching assistants
- Created and graded assignments, exams, and class projects
- Designed and presented lecture materials
- Supported students during office hours and class forums

**Project Mentor, Dept. of Computer Science,** Carnegie Mellon University *09/20-Present*
- Designed a term project for 15-821 (Mobile and Pervasive Computing)
- Met and provided guidance to project members on a regular basis

## Publications (Reverse Chronological Order)

[1] **Haithem Turki**, Michael Zollhoefer, Christian Richardt, Deva Ramanan, "PyNeRF: Pyramidal Neural Radiance Fields." To appear in Neurips 2023.

[2] Shilpa George, **Haithem Turki**, Ziqiang Feng, Thomas Eiszler, Deva Ramanan, Padmanabhan Pillai, Mahadev Satyanarayanan, "Low-Bandwidth Self-Improving Transmission of Rare Training Data." In Proceedings of the 29th Annual International Conference on Mobile Computing And Networking (MobiCom '23).

[3] Shilpa George, **Haithem Turki**, Ziqiang Feng, Thomas Eiszler, Deva Ramanan, Padmanabhan Pillai, Mahadev Satyanarayanan, "Edge-based Privacy-Sensitive Live Learning for Discovery of Training Data." In Proceedings of the 1st International Workshop on Networked AI Systems (NetAISys '23).

[4] **Haithem Turki**, Jason Y. Zhang, Francesco Ferroni, Deva Ramanan. "SUDS: Scalable Urban Dynamic Scenes." In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023.

[5] Mahadev Satyanarayanan, Ziqiang Feng, Shilpa George, Jan Harkes, Roger Iyengar, **Haithem Turki**, Padmanabhan Pillai, "Accelerating Silent Witness Storage," in IEEE Micro, Nov.-Dec. 2022.

[6] **Haithem Turki**, Deva Ramanan, Mahadev Satyanarayanan, "Mega-NeRF: Scalable Construction of Large-Scale NeRFs for Virtual Fly-Throughs." In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022.

[7] Shilpa George, Thomas Eiszler, Roger Iyengar, **Haithem Turki**, Ziqiang Feng, Junjue Wang, Padmanabhan Pillai, Mahadev Satyanarayanan, "OpenRTiST: End-to-End Benchmarking for Edge Computing," in IEEE Pervasive Computing, vol. 19, no. 4, pp. 10-18, 1 Oct.-Dec. 2020

[8] Mahadev Satyanarayanan, Thomas Eiszler, Jan Harkes, **Haithem Turki** and Ziqiang Feng, "Edge Computing for Legacy Applications," in IEEE Pervasive Computing, 1 Oct.-Dec. 2020

[9] M. Pawan Kumar, **Haithem Turki**, Dan Preston and Daphne Koller, "Parameter Estimation and Energy Minimization for Region-Based Semantic Segmentation," in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 37, no. 7, pp. 1373-1386, 1 July 2015.

[10] M. Pawan Kumar, **Haithem Turki**, Dan Preston and Daphne Koller, "Learning specific-class segmentation from diverse data," 2011 International Conference on Computer Vision.

## Patents

[1] **Haithem Turki,** Robert Fink, Amr Al Mallah. "Systems and Methods for Indexing and Searching." US Patent 2019/0347343 A1, filed June 8, 2018, and issued November 14, 2019.

[2] **Haithem Turki,** Sander Kromwijk, Stephen Cohen, Yixun Xu, Feridun Arda Kara. "Systems and Methods for Constraint Driven Database Searching." US Patent 2018/293239 A1, filed April 11, 2017, and issued October 11, 2018.

## Technical Reports

[1] Ziqiang Feng, Shilpa George, **Haithem Turki,** Roger Iyengar, Padmanabhan Pillai, Jan Harkes, Mahadev Satyanarayanan. "Improving Edge Elasticity via Decode Offload." CMU-CS-21-139, September 2021.

## Additional Information

- **Conference Reviewer:** CVPR, ICCV, SIGGRAPH Asia
- **Volunteer Activities:** Alchemist Accelerator (mentor and investor)
- **Languages:** Fluent in French and Arabic. Currently learning Japanese and German.
- **Other Qualifications:** Eberly Future Faculty Program. Top Secret security clearance (current). FINRA Series 65. Graduate of the Stanford StartX startup accelerator.